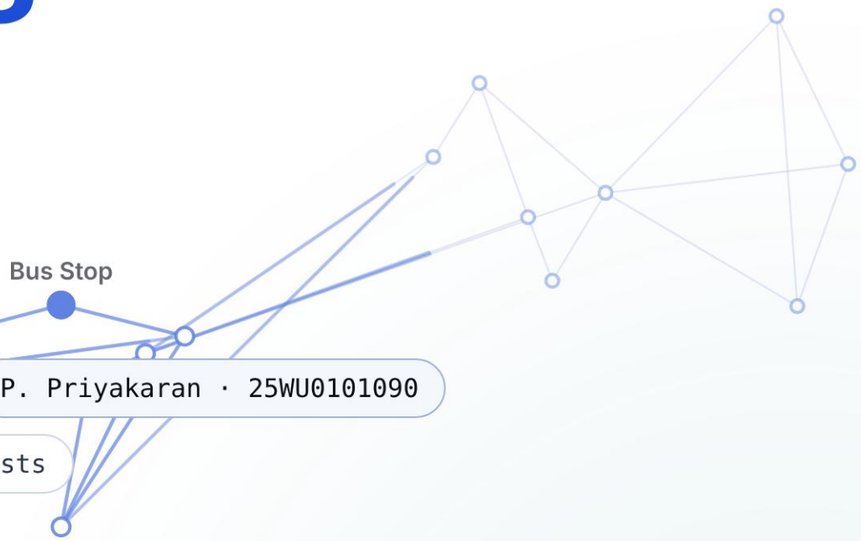
— DESIGN AND ANALYSIS OF ALGORITHMS · FINAL REVIEW

Campus Shuttle Routing & Capacity Scheduling

One pipeline, five algorithm families: cutting the average passenger wait across 14 Woxsen campus stops.



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Group project · 202 tests

— 01 · PROBLEM

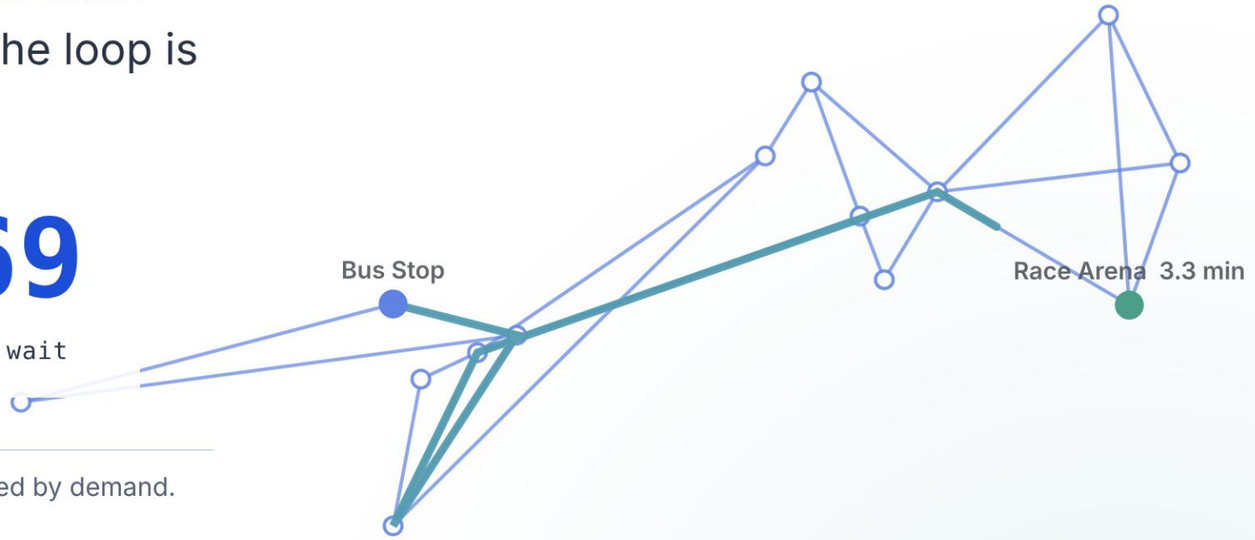
A single shuttle in fixed stop order makes people wait

14 stops from Main Gate to SportX. Visiting them in list order sends the shuttle back and forth across campus, so the loop is long and every stop waits half of it on average.

14
stops with real coordinates

13.37
min loop in file order

6.69
min average wait (baseline)



Average wait = loop time / 2, assuming passengers arrive uniformly and stops are weighted by demand.

— 02 · APPROACH

Model the campus as a weighted graph, then solve it in layers

GRAPH

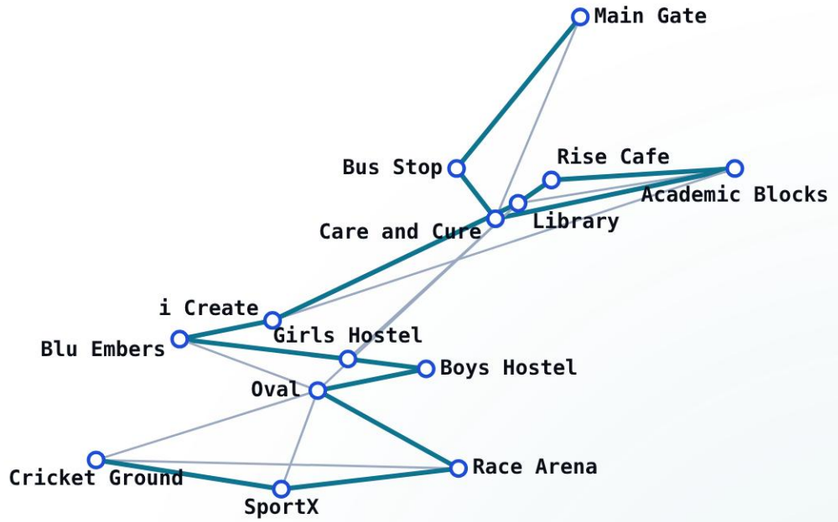
14 vertices, 22 edges

Edge weight = haversine distance between real stop coordinates at an assumed 20 km/h.

ALGORITHMS

MST · SSSP · APSP · Greedy · DP · TSP

Each one answers a different operational question, and each is cross-checked against a second method.



Cyan = minimum spanning tree (5.8 min).

— 03 · GRAPH ALGORITHMS

Three ways to look at the road network, all cross-verified

MST

Prim and Kruskal

Cheapest set of roads connecting every stop. Both give the same tree.

5.8 min

Prim = Kruskal

SSSP

Dijkstra and Bellman-Ford

Fastest route Bus Stop to Race Arena via Care and Cure, Academic Blocks, Library, Oval.

3.3 min

no negative cycle

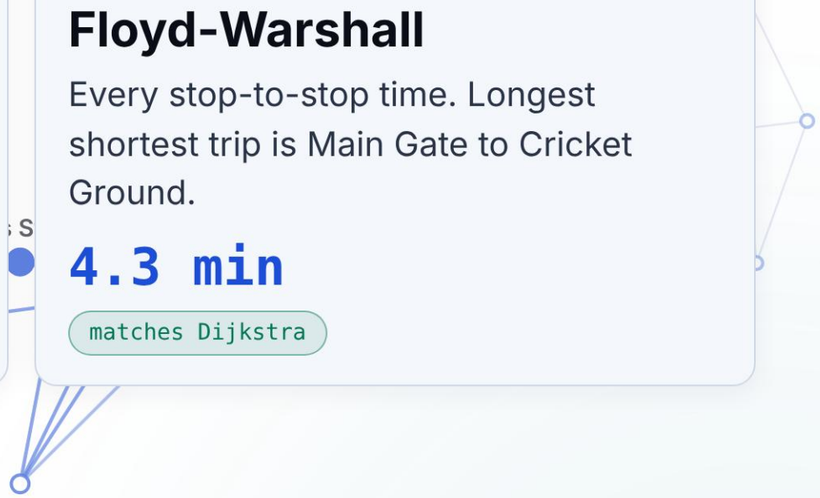
APSP

Floyd-Warshall

Every stop-to-stop time. Longest shortest trip is Main Gate to Cricket Ground.

4.3 min

matches Dijkstra



— 04 · SCHEDULING AND ALLOCATION

Greedy staffs the shifts, DP fills the vehicles

GREEDY · INTERVAL SCHEDULING

Earliest-finish-first

One driver can cover 8 of 18 shifts. Interval partitioning shows 4 drivers staff all shifts.

8 / 18 · 4 drivers

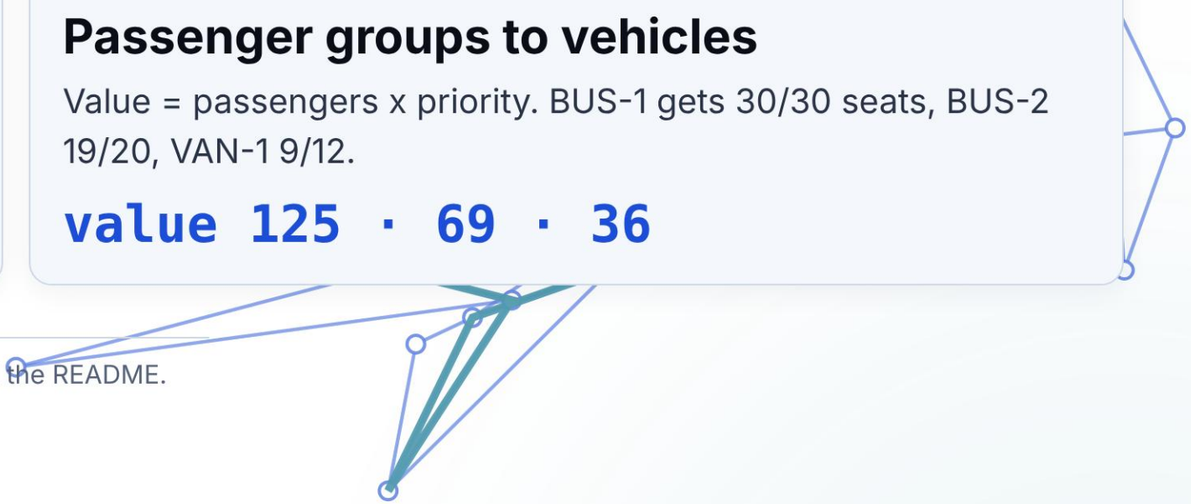
DP · 0-1 KNAPSACK

Passenger groups to vehicles

Value = passengers x priority. BUS-1 gets 30/30 seats, BUS-2 19/20, VAN-1 9/12.

value 125 · 69 · 36

nd 3.3 min



Demand, fleet and shift data are placeholders, labelled ESTIMATE in the CSV headers and the README.

— 05 · TOUR OPTIMISATION

Exact TSP proves the heuristics are within 2% of optimal



1.019x
MST tour vs optimal

≤ 2x
theoretical MST bound



Held-Karp is $O(n^2 2^n)$, exact and feasible at n = 14. MST preorder is O(V log V) and scales to much larger campuses.

— 06 · RESULT

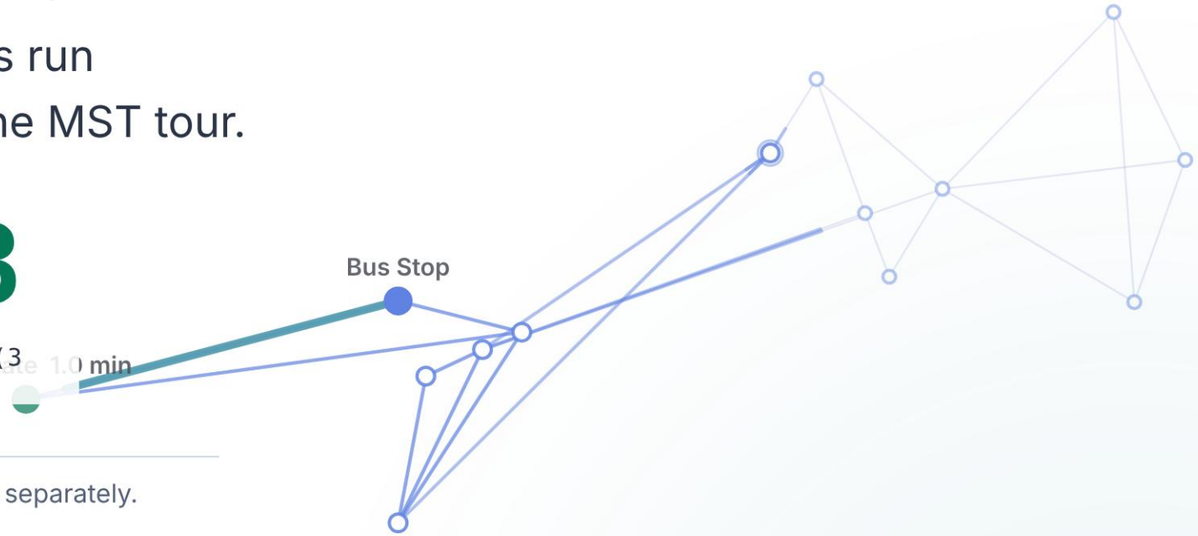
Average wait cut by 70%

Baseline: one shuttle in file order. After: three shuttles run concurrently, each looping one contiguous slice of the MST tour.

6.69
min wait before

5.05
min, 1 shuttle on MST tour

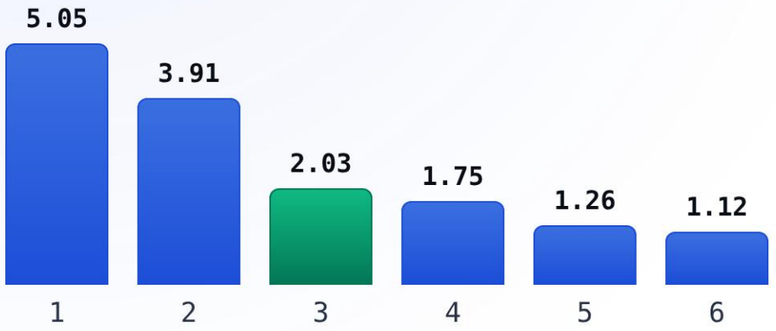
2.03
min wait after (3 shuttles)



Routing alone saves 1.6 min; adding fleet capacity saves the rest. Both effects are shown separately.

— 07 · FLEET SENSITIVITY

Three shuttles is the knee of the curve



Average wait (min) by number of shuttles

Going from 2 to 3 shuttles saves 1.88 min. The 4th saves only 0.28. Improvement per extra shuttle first drops below 0.5 min at 4 shuttles, so 3 shuttles (the size of the estimated fleet file) is the sensible operating point.

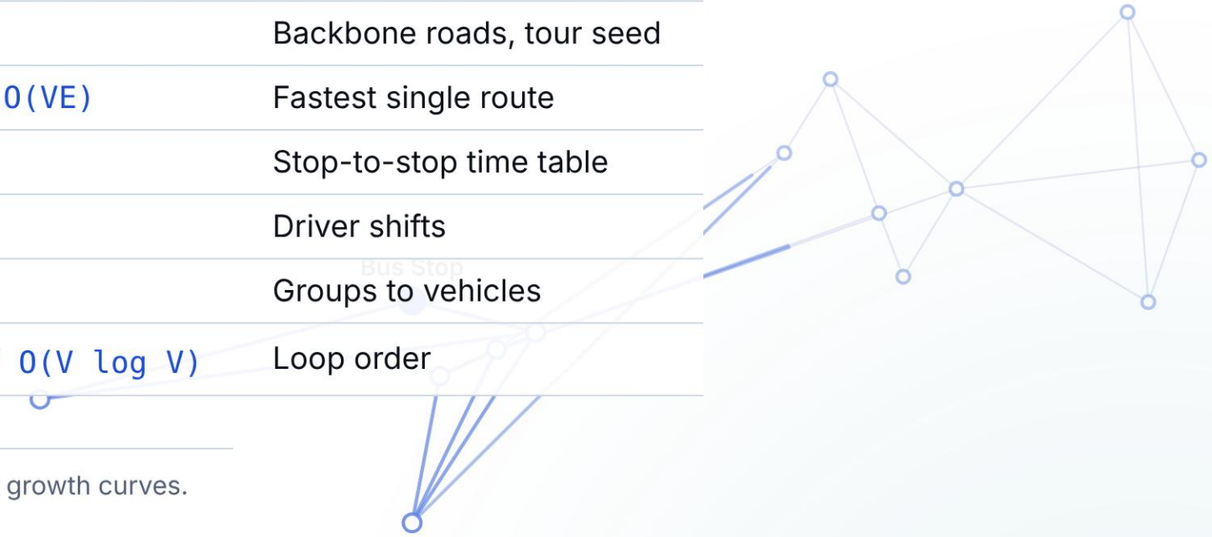
Main Gate 1.0 min

— 08 · COMPLEXITY

Every algorithm, its cost and why it fits

PROBLEM	ALGORITHM	TIME	USED FOR
MST	Prim / Kruskal	$O(E \log V)$	Backbone roads, tour seed
SSSP	Dijkstra / Bellman-Ford	$O((V+E) \log V)$ / $O(VE)$	Fastest single route
APSP	Floyd-Warshall	$O(V^3)$	Stop-to-stop time table
Scheduling	Greedy (earliest finish)	$O(S \log S)$	Driver shifts
Allocation	0-1 knapsack DP	$O(nW)$	Groups to vehicles
Tour	Held-Karp / NN / MST preorder	$O(n^2 2^n)$ / $O(n^2)$ / $O(V \log V)$	Loop order

A runtime-vs-input-size benchmark (50 to 1600 nodes) in the report folder confirms the growth curves.



— 09 · VERIFICATION

Tested, cross-checked and honest about its data

TESTING

202 tests pass

Algorithms compared against each other: Prim vs Kruskal, Dijkstra vs Bellman-Ford vs Floyd-Warshall, Held-Karp and knapsack vs brute force.

REAL DATA

14 surveyed stops

Names and GPS coordinates are real. Travel times derive from them.

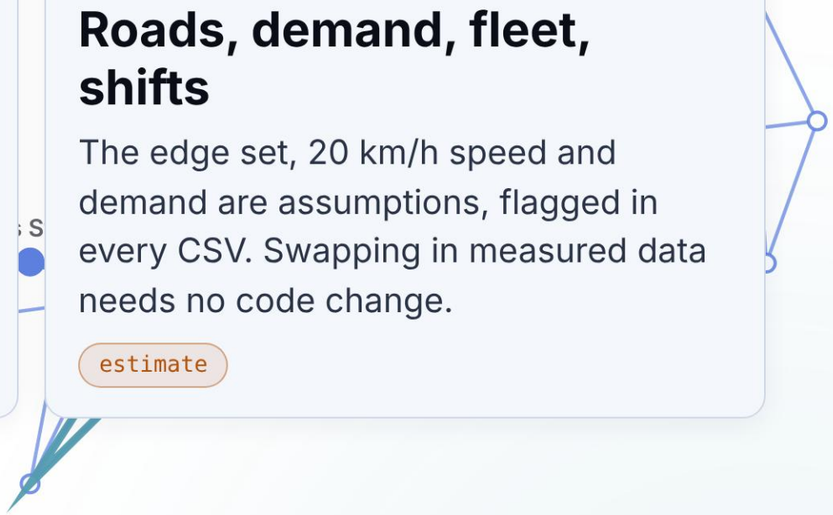
real

ESTIMATED DATA

Roads, demand, fleet, shifts

The edge set, 20 km/h speed and demand are assumptions, flagged in every CSV. Swapping in measured data needs no code change.

estimate



— 10 · DELIVERABLES

Everything runs with one command

RUN

`python main.py`

Runs every algorithm, benchmarks, and regenerates the report folder.

OUTPUTS

Interactive dashboard + charts

dashboard.html, runtime_vs_complexity.png and fleet_sensitivity.png.

FUTURE WORK

Make it measured

Survey real roads and ride times, count real boardings, add dwell time and road-detour factors, then rerun unchanged.



— THANK YOU

Questions?

Campus Shuttle Routing & Capacity Scheduling. Design and Analysis of Algorithms, final review.

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